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ABSTRACT

The near-cathode region plays a crucial role in exploring the transport characteristics of the transition from arc column to the hot cathode in atmospheric-pressure arc discharges because of the existing non-equilibrium phenomena. A one-dimensional unified model, including the near-cathode region and the cathode body, is developed for an argon arc discharge with the tungsten cathode at atmospheric pressure in this paper. The electrostatic model coupled with an external circuit in the near-cathode region is solved based on the implicit particle-in-cell coupled Monte Carlo collision method without any assumptions of thermal or ionization equilibrium or quasi-neutrality. A detailed description of the arc plasma–cathode and cathode–gas interactions is obtained by calculating the nonlinear heat conduction equation in the cathode. It is shown that the space-charge sheath strongly affects particle transport in the near-cathode region and energy transport from arc plasma to the thermionic cathode. The total current density has significant effects on the kinetic characteristics of arc plasma by feedback-like mechanisms. The Joule heating by the external circuit and charged particles deposited into the cathode are dominating mechanisms of energy transfer from the near-cathode region to the cathode, while energy loss by radiation is more significant compared with natural convection.

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I. INTRODUCTION

Atmospheric-pressure arc discharge can generate high-intensity thermal plasma, which is widely applied in various industrial fields as a heat source, such as welding, waste treatment, and electric arc furnace steelmaking.[1–8] Published studies show that the interaction between the thermal plasma and the electrode surface plays an important role in the generation and maintenance of arc discharge.[9–11] The arc plasma in the near-cathode region deviates from the local thermodynamic and chemical equilibrium with strong nonlinear characteristics, involving the multi-physical coupling effect of electromagnetic field, flow field, and non-uniform temperature field.[12] With the increasing demand for effective control of thermal plasma technology, adequate descriptions of the near-cathode region and a deep understanding of the plasma–cathode interaction are the key elements in exploring atmospheric-pressure arc discharges.

Due to the small spatial scale and large parameter gradients, the near-cathode region of typical arc discharges represents an extremely difficult object for experimental investigation. The experiments cannot provide much guidance to study the transport characteristics of thermal plasma near the cathode. Therefore, the numerical simulation methods become alternative and powerful tools to investigate the complicated features of near-cathode plasma regions.[13–24] Most of these studies divided the near-cathode region into several sub-layers according to different properties, such as space-charge sheath, Knudsen layer, and ionization layer. Each sub-layer was described by respective governing equations and boundaries between adjacent sub-layers were determined based on conservation conditions. However, the division of sub-layers involves quite a bit of intuitive consideration and, therefore, is not a commonly accepted physical method.[25]

An alternative is to model the whole near-cathode region in the framework of a single set of governing equations[25–29] in which there was no regional division and the assumptions, including quasi-neutrality, thermal equilibrium, and chemical equilibrium, were adopted. Based on this model, physical mechanisms dominating different regions can be identified and sub-layers can be pinned down appropriately.[25] A two-dimensional model of high-intensity arcs was reported in Ref. 26 to study the effects of the plasma's gas flow rate on the negative anode fall and the heat flux to the anode. Moreover, a two-temperature hydrodynamic model, including the arc column, the anode, and the cathode, was developed in Refs. 11 and 27 for

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high-current free-burning arcs, analyzing the energy transport between the arc column and the cathode and the characteristics of deviating from local chemical equilibrium (LCE) in the near-cathode region. On the other hand, a one-dimensional (1D) model[28,29] was performed in the framework of hydrodynamics transport equations for the ions and the electrons. The diffuse current transfer in the near-cathode region was calculated for the high-pressure molecular plasma. However, it was available only for conditions of low current densities and low ionization degree. Since the thickness of the near-cathode region is much smaller than the transversal dimension of the cathode, variations of plasma parameters across the radial cross section are regarded to be much faster than those in transversal directions.[14] Thus, 1D model is reasonable and has been widely applied in most existing investigations on the near-cathode regions in high-pressure arc discharges.

Most of the above-described approaches assume that the space-charge sheath is collision-free and the particle velocity distribution function satisfies the Maxwell distribution. However, the distribution function of electrons and ions may deviate from the Maxwell distribution in sheath. In addition, Knudsen number $\mathrm{Kn} = \frac{\lambda}{L} > 0.1$ (λ: particle mean free path; L: characteristic length) in this region.[30] Therefore, the traditional fluid dynamics methods are no longer applicable.

In this work, 1D unified model is developed for the atmospheric pressure arc discharge driven by a current source concerning the interactions of arc–cathode and cathode–gas to fully explain the particle and energy transport phenomenon and kinetic characteristics of arc plasma. The electrostatic model in the near-cathode region is solved based on the implicit particle-in-cell (PIC) coupled with the Monte Carlo Collision (MCC) method without any assumptions of thermal or ionization equilibrium or quasi-neutrality, coupled with the nonlinear thermal conduction problem of the cathode model. The outline of the paper is as follows. In Sec. II, governing equations of this model are given, together with corresponding boundary conditions. Dominating physical mechanisms are identified and the validity of assumptions is analyzed on which available simplified models are based. Numerical simulation results are given and discussed in Sec. III. The distribution of electric potential, electric field, particle number density, and spatially averaged temperature, as well as particle energy distribution function in the near-cathode region, are predicted, and the temperature distribution of the cathode body is obtained self-consistently by using the above developed kinetic model. The densities of the energy flux used to heat and cool the cathode are analyzed and mechanisms causing the non-monotony of the dependence of the energy flux density on the surface temperature are elucidated. The effects of different total current densities on plasma characteristics are also evaluated analytically. Finally, the main conclusions are summarized in Sec. IV.

II. METHODOLOGY

A. Physical model

In 2008, Almeida *et al.* proposed 1D unified modeling of near-cathode plasma layers in a high-pressure mercury arc and an atmospheric-pressure argon arc discharges.[25] Here, the cathode body and part of the near-cathode region (i.e., the cathode sheath and part of the pre-sheath) are selected as the computational domain to study the plasma–cathode interaction and the particle transport phenomenon in the near-cathode region. Figure 1 shows the schematics of arc discharge [Fig. 1(a)] and the calculation domain [Fig. 1(b)]. Biot number is $Bi = \frac{h_s R}{\lambda}$ (h_s: surface heat transfer coefficient; R: radius of the cathode; λ: thermal conductivity of cathode) with the diameter of the tungsten rod of 1 mm in this work. Therefore, the sectional thermal resistance can be negligible compared to surface convective thermal resistance. At this point, the cathode temperature in the cross section is considered to be evenly distributed. As previously stated, it is reasonable to apply a 1D model to investigate the kinetic characteristics of the near-cathode regions for high-pressure arc discharges. Thus, the 1D cathode model and 1D near-cathode model are adopted in the following calculations. The components of particle flux and particle collision processes in the near-cathode region and heat flux crossing the cathode are shown in Fig. 1(b). A constant current-driven method ($j_{\mathrm{total}} = \mathrm{const.}$) is used, which is typical for high-pressure DC arc discharges.

Benilov[12] and Almeida *et al.*[25] have proved that it is located in the pre-sheath region with a distance away from the cathode about $\sim 10^{-6}$ m, and the arc plasma in this region meets the characteristics of local chemical equilibrium. Therefore, the length of the near-cathode region is 5.0 μm in this work and the length of the cathode body is 14 mm according to Ref. 11. Argon is used as the plasma forming gas and the surrounding gas under high pressure of 1 atm, and the cathode material is refractory tungsten.

Three kinds of particles are comprised in the arc plasma, i.e., argon atoms (Ar), ions (Ar^+), and free electrons (e); the presence of multiply charged ions is neglected as justified in Ref. 25. The frequency of Coulomb collision is relatively low, so only the collisions between charged particles and neutral atoms are considered in this work, i.e., elastic collision, excitation, and ionization between e and Ar; elastic collision and charge exchange between Ar^+ and Ar. Since there are many electron energy levels and excited states for metastable argon atoms, the more important excited states are taken into consideration in this work, i.e., the first excited state with the first excitation potential of 11.5 eV. Generally, the electron impact ionization process leads to the exponential increase in electrons in the volume. Charged particles transport through drift and diffusion in the near-cathode region, and part of the particles are absorbed by the cathode. Meanwhile, most electrons from the cathode emission drift to the arc column and are trapped in the anode. To sustain arc discharge, these electrons must be added through the electron emission from the cathode. Our previous study in Ref. 32 has proved that thermionic emission is the dominant electron emission process at the cathode surface for the atmospheric arc discharges. The influence of secondary electron emission, which often occurs in the glow discharges and field electron emission usually in the atmospheric microdischarges or vacuum arc discharges, can be ignored. Hence, the emitted electrons through thermionic emission are accelerated by the electric field. The cathode can be heated by charged particles reaching the cathode surface and loses energy by electron emission, convection, and radiation with the surrounding. The surrounding gas is at room temperature, which is called cold gas in this study. It is believed that the steady-state temperature distribution within the cathode body can be obtained when the discharge reaches stability. In what follows, this computational domain is described by employing a whole model with proper coupling conditions, i.e., arc plasma–cathode and cathode–cold gas.

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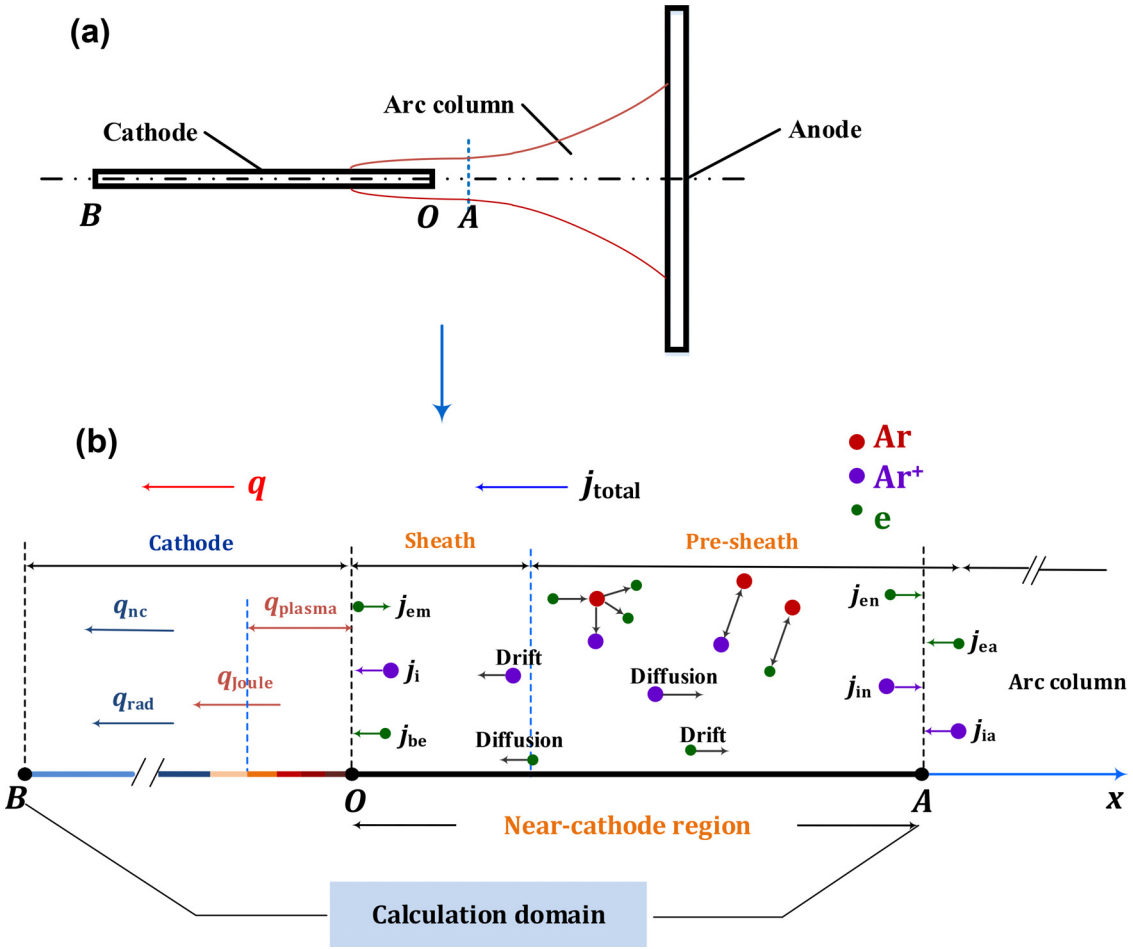


FIG. 1. Schematic of arc discharge (a) and the calculation domain (b), including cathode and near-cathode region.

B. Mathematical models

1. Electrostatic model coupled with an external circuit

Since the wavelength of the electromagnetic wave is much larger than the size of the discharge device, the self-induced magnetic field generated by arc plasma can be ignored. Therefore, an electrostatic model is used to calculate the electric field and the surface charge of the cathode by solving the Poisson equations,

$$\varepsilon \frac{d^2\varphi}{dx^2} = e(n_e - n_i), \tag{1}$$

where φ is electric potential, ε is the vacuum dielectric constant, and n_e and n_i are the number density of electrons and ions, respectively. The potential, electric field, and charge density are defined on the grid node to avoid the nonphysical effects of the electrostatic model. $E = -d\varphi/dx$, so the central difference scheme is applied in the calculation of the electric field. The near-cathode region is divided into N grid cells evenly, as seen in Fig. 2.

Since the electric potential of the pre-sheath region is extremely small, it is approximately zero for a one-dimensional flat panel system.[31] Therefore, the boundary conditions are defined as

$$\varphi_N = 0, \tag{2}$$

$$E_0 = \sigma_0/\varepsilon, \tag{3}$$

where σ_0 is the charge density on the cathode with units of charge/area and E_0 is the electric field intensity at the cathode. According to Gauss's law, the electric field at one-half grid cell from the cathode boundary $E_{1/2}$ is[31]

$$E_{1/2} = \frac{\varphi_0 - \varphi_1}{\Delta x} = \frac{1}{\varepsilon}\left(\sigma_0 + \rho_0 \frac{\Delta x}{2}\right), \tag{4}$$

where Δx is the grid step and ρ is total charge per unit volume.

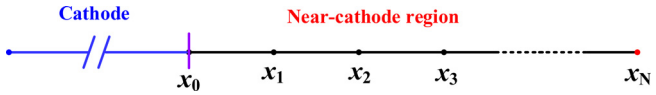


FIG. 2. Locations of grid points relative to the boundary.

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Bringing the boundary conditions into the discrete Poisson equation, the three-diagonal matrix form can be written as follows:

$$\begin{bmatrix} b_0 & c_0 & 0 & \cdots & \cdots & & \\ a_1 & b_1 & c_1 & 0 & & & \\ 0 & a_2 & b_2 & c_2 & 0 & \cdots & \\ \vdots & & & \vdots & & & \cdots \\ & \vdots & & \vdots & \vdots & & \\ \vdots & & & \vdots & \vdots & & \\ & \vdots & & a_{N-2} & b_{N-2} & c_{N-2} & \\ & & & & a_{N-1} & b_{N-1} & \end{bmatrix} \times \begin{bmatrix} \varphi_0 \\ \varphi_1 \\ \varphi_2 \\ \vdots \\ \vdots \\ \vdots \\ \varphi_{N-2} \\ \varphi_{N-1} \end{bmatrix}^t = f \begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ \vdots \\ \vdots \\ \vdots \\ d_{N-2} \\ d_{N-1} \end{bmatrix}^t. \tag{5}$$

The superscript indicates the parameter calculated at t. The matrix elements are

$$\begin{gathered} a_n = 1, \quad n = 1, 2, \ldots, N-1, \\ b_0 = -1, \quad b_n = -2, \quad n = 1, 2, \ldots, N-1, \\ c_n = 1, \quad n = 0, 2, \ldots, N-2, \end{gathered} \tag{6}$$

$$d_0 = \frac{\sigma_0^t}{\Delta x} + \frac{\rho_0^t}{2}, \quad d_n = \rho_n^t, \quad n = 1, 2, \ldots, N-1, \tag{7}$$

$$f = -\Delta x/\varepsilon. \tag{8}$$

The external current-driven circuit is coupled to Eqs. (1)–(8) through charge conservation at the cathode,

$$A\Delta\sigma = Q_{\text{conv}} + \Delta Q, \tag{9}$$

where A is the cross section of the cathode, Q_{conv} is the total charge of the charged particles absorbed into the cathode, and ΔQ is the charge deposited by the external circuit current. Hence, the surface charge of the cathode is updated during every iteration and remains approximately constant when the calculation is stable.

2. Current conservation model

When exploring the process that different current transfers to the refractory cathode, the effects of metal vapor from the cathode surface on the kinetic characteristics can be ignored. An ideal current-driven method is assumed in the present work. The external circuit elements can be ignored since an ideal current source can be treated as an open circuit. The current density at the cathode j_{CA} is mainly composed of ions and electrons from the sheath by drift and diffusion and thermionic emission electrons. The expression[32] is as follows:

$$j_{\text{CA}} = -j_{\text{em}} - j_{\text{i}} + j_{\text{be}}, \tag{10}$$

$$j_{\text{em}} = j_{\text{TE}}(T_{\text{C}}, E_{\text{C}}) = A_{\text{RD}} T_{\text{C}}^2 \exp\left(-\frac{e\phi - \sqrt{e^3 E_C/4\pi\varepsilon}}{k_{\text{B}} T_C}\right), \tag{11}$$

where j_{em} is thermionic emission current density described by the Schottky equation for modest electric fields, j_{be} represents the back diffusion current density of highenergy electrons, and j_{i} is due to the bombardment of ions arriving at the cathode surface. A_{RD} is the Richardson coefficient of 120 A cm^{-2} K^2 and ϕ is the work function of 4.55 eV, respectively (employed in these studies[32–34]) k_{B} is Boltzmann's constant and T_{C} is the cathode temperature.

The current density at the right boundary in the near-cathode region j_{Right}[32] is

$$j_{\text{Right}} = j_{\text{in}} + j_{\text{ea}} - j_{\text{ia}} - j_{\text{en}}, \tag{12}$$

where j_{in} is the ion current density moving toward the right boundary due to the ion diffusion, j_{en} is the electron current density arriving at the right boundary driven by the electric field in the near-cathode region, j_{ia} is formed by the ion flux accelerated by the electric field and directed to the cathode surface, j_{ea} is the electron current density entering the near-cathode region from the arc column by diffusion, which is usually very small compared with j_{ia}.

In terms of current-driven arc discharges, the conservation of the current density should be satisfied physically at two boundaries during every iteration step of the calculation process. According to Kirchhoff's current law at the cathode boundary, if $j_{\text{total}} > j_{\text{CA}}$, we assume that the exceeding current is carried by the ions, while if $j_{\text{total}} < j_{\text{CA}}$, the exceeding current is carried by the electrons and is assumed to be the diffusion current. Similarly, the difference between j_{total} and j_{Right} is determined as a diffusion current entering the pre-sheath from the arc column. The velocity and temperature of supplementary charged particles on the right boundary (seen in Fig. 1) are determined based on the physical properties of arc plasma in the pre-sheath (i.e., local chemical equilibrium). Subsequently, the velocity and temperature of supplementary charged particles at the cathode surface are determined based on mass and energy conservation. Then, Eqs. (1)–(12) are solved based on the charge density obtained from the finite difference of Eq. (9) to obtain the kinetic characteristics of arc plasma in the near-cathode region.

3. Energy fluxes between near-cathode region and cathode

The ions, after having been accelerated by the electric field in the sheath, bring their energy to the cathode surface. Moreover, additional heating is provided by fast electrons that can overcome the decelerating electric field in the sheath and reach the cathode surface. Joule heat provided by an external circuit should be taken into consideration. These are three major mechanisms of heating the cathode. Moreover, the cathode is cooled by the effects of convection and radiation with the cold surrounding. The governing equation of the heat conduction in the cathode body is

$$\frac{d}{dx}\left(\lambda(T_C)\frac{dT_C}{dx}\right) + \dot{\Phi} = 0, \tag{13}$$

where T_{C} is the temperature of the cathode, $\dot{\Phi}$ is the volumetric heat source term over the entire cross section converted by the heat exchanged through the cathode surface with the unit of W/m^3 and derived later, and $\lambda(T_C)$[35,36] is the thermal conductivity of the cathode given by fitting the results of the experiment by Allen *et al.*,[36] that is, $\lambda(T_C) = 12331\, T_C^{-0.738}$.

At the left boundary, we assume that the temperature is constant because the cathode surface is externally cooled. Thus,

$$x = -L, T_C = 300\,\text{K}. \tag{14}$$

At the right boundary, we have

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$$x = 0, \quad -\lambda(T_C)\frac{dT_C}{dx} = q_{\text{plasma}}(T_C). \tag{15}$$

Here, q_{plasma} is the net heat flux of arc plasma deposited into the cathode surface and L is the length of cathode.

In the framework of this model, heat generation by Joule heat and net particle deposition and energy loss by natural convection and radiation are taken into consideration. Then, the volumetric heat source term is given by

$$\dot{\Phi} = P_{Joule} + \frac{(q_{plasma} - q_{rad} - q_{nc})C}{A}, \tag{16}$$

$$P_{\text{Joule}} = j_{\text{total}}^2 \rho_l(T_C), \quad q_{\text{nc}} = h_{nc}(T_C - T_g), \quad q_{\text{rad}} = \varepsilon(T_C)\sigma\left(T_C^4 - T_g^4\right), \tag{17}$$

where P_{Joule} is the volumetric power density with a unit of W/m^3; q_{nc} and q_{rad} are the heat flux of natural convection and radiation with units of W/m^2, respectively; $\rho_l(T_C)$ is the resistivity with a unit of Ω m given by $\rho_l(T_C) = 0.0489 \times 10^{-8}(1+0.006\,03 \times (T_C - 273))$; the radiant emissivity $\varepsilon(T_C)$ is given by $\varepsilon(T_C) = 0.000\,108 \times T_C + 0.0224$; C is the cross-sectional perimeter of the cathode; h_{nc} is the coefficient of natural convection heat transfer of 10 W/(m^2 K) in this work; σ is Stefan–Boltzmann constant of 5.67×10 W/(m^2 K^4); and T_{g}, the temperature of ambient gas, is 300 K.

The density of the net energy flux to the cathode surface q_{plasma} is given by the following expression:[31]

$$q_{\text{plasma}} = q_{\text{i}} + q_{\text{e}} - q_{\text{em}}, \tag{18}$$

$$q_{\text{em}} = \frac{j_{\text{em}}}{e}(2k_B T_{\text{C}} + \phi). \tag{19}$$

Here, q_{i} and q_{e} represent the densities of energy fluxes delivered to the cathode surface by ions and by electrons, respectively, and q_{em} is the density of energy losses due to thermionic emission. The units of these parameters are W/m^2.

According to the results presented in Ref. 37, there is no thermionic emission from the cathode if the cathode temperature is too low. That is, Eqs. (11) and (19) are not applicable when T_{C} is below a certain temperature. The solution used in this work refers to Ref. 12, in which 2000 K is chosen to be a certain temperature. Therefore, q_{em} and q_{plasma} are set to be zero when $T_{\text{C}} < 2000$ K.

In the framework of the above approach, a description of the arc plasma–cathode interaction may be constructed in the following three steps. First, kinetic parameters of arc plasma in the near-cathode region are obtained by solving Eqs. (1)–(12) by means of the PIC-MCC method. Second, densities of the energy flux and the current flux from the plasma to the cathode surface are statistically given based on Eqs. (17)–(19). Third, the nonlinear heat conduction problem [Eqs. (13)–(16)] is solved numerically to obtain the temperature distributions of the cathode body.

C. PIC-MCC method

The PIC-MCC model provides a more effective method for studying gas discharges by using a certain number of simulation particles instead of real particles. The simulation particles have the same charge-to-mass ratio, coordinates, and speeds as real particles. The PIC model is employed to describe the particle motion, and the MCC model accounts for the collision processes.

For the PIC model, it is necessary to distinguish the smallest and fastest physical process with suitable space steps and time steps. In addition, the simulation needs to execute until the slowest physical process in the system reaches equilibrium. By introducing an appropriate implicit scheme to the particle simulation,[32,33,38–40] the limitations of the electronic oscillation frequency and Debye length can be eliminated. Thus, a larger space step can be used in the calculation, thereby greatly reducing the computing effort.

For the implicit PIC-MCC model, stability constraints can be expressed as

$$\Delta t < \frac{\Delta x}{v_t} = \frac{\Delta x}{\sqrt{\frac{8kT_e}{\pi m_e}}}, \tag{20}$$

where v_{t} is the thermal velocity of electrons and the calculation is stable only if most particles move less than two cells within one time step. Time step Δt is fixed at 10^{-13} s in this work.

Another issue is that different weights of simulation particles might affect calculation results, so the particle rezoning method by Lapenta is adopted to eliminate this problem.[41] In terms of atmospheric pressure arc discharge, 2000 simulation particles are placed in each cell. A particle merging algorithm is used when the number of simulation particles exceeds 2000 per cell.

D. Grid independence check

The grid independence check is essential in numerical simulations. Here, the cathode body is uniformly divided into 1000, 1500, and 2000 cells, and the calculated temperature at $x = 0$ using these three sets of space steps is individually 3260.85, 3260.85, and 3260.86 K with little relative discrepancy. Meanwhile, six sets of grid systems (i.e., 32, 64, 128, 192, 256, and 512 cells, respectively) are applied in the near-cathode region referring to Refs. 32, 33, and 38–40. The corresponding time steps are determined under the premise of satisfying the stability of the PIC-MCC method. Figure 3(a) shows that the values of electric field at $x = 0$, E_0, vary with the cell numbers. Since the direction of the electric field is opposite to that of the x-axis, E_0 is negative. The calculated E_0 are -4.0712×10^7, -4.0772×10^7, and -4.077×10^7 V m^{-1} for 192, 256, and 512 cells, respectively. It can be seen that the relative discrepancy of electric fields between 256 and 512 is within 2%, which means that grid independence has been achieved. Meanwhile, the results of iterative convergence are given. Figures 3(b) and 3(c) present the distribution of electric field in the near-cathode region and charge density at $x = 0$ σ_0 for different iteration times. There is almost no change in the distribution of electric field for different iteration times (10^5, 10^6, 5×10^6) as shown in Fig. 3(b). However, Fig. 3(c) shows that the charge density presents tiny variations after running over 5×10^6 time steps. Considering the error of the statistical average, it is thought that the simulation is convergent.

So to summarize, considering the calculation cycle and calculation accuracy, the cathode and the near-cathode region are uniformly divided into 1000 and 256 cells, respectively, while the time step is fixed at 1×10^{-13} s and the simulations run for 5×10^6 time steps in the following simulation.

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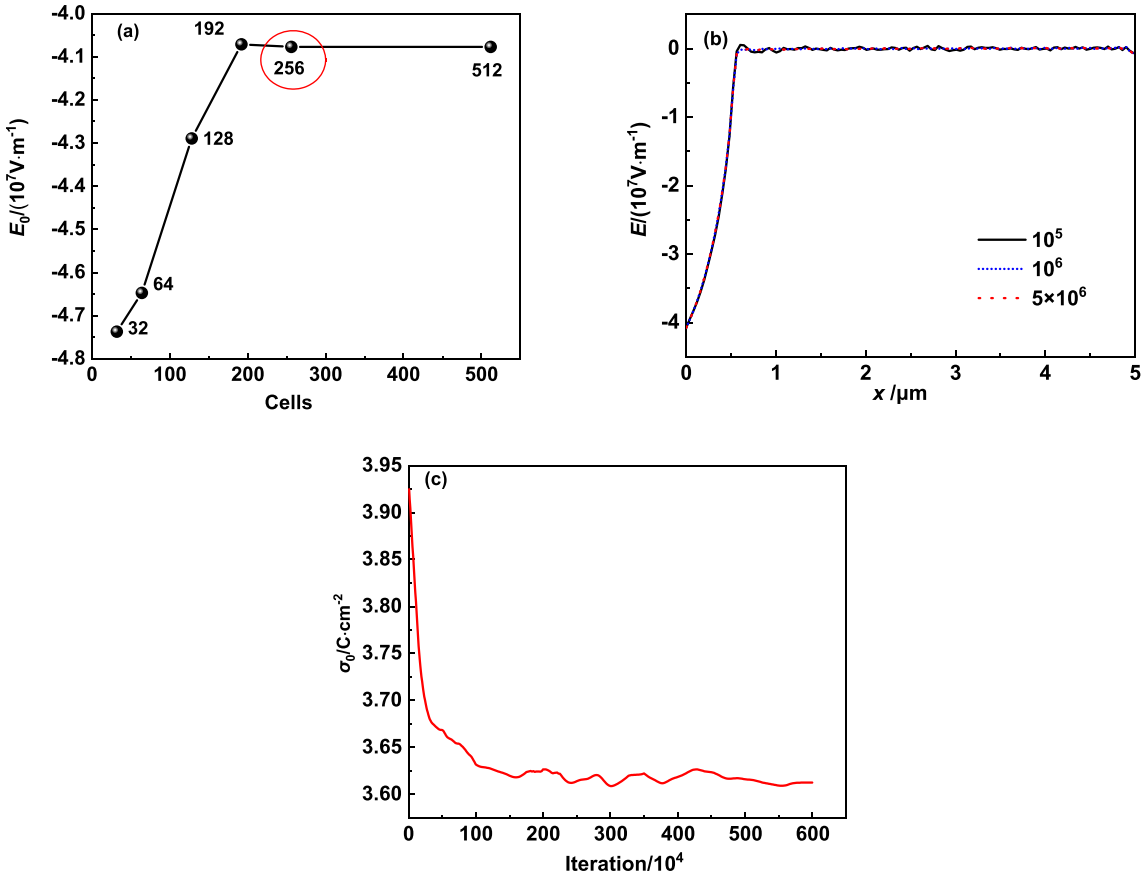


FIG. 3. Grid independence check. (a) Electric field at $x = 0$ E_0 vs cells. (b) Distribution of the electric field in the near-cathode region vs iteration times. (c) Charge density σ_0 vs iteration times.

III. RESULTS AND DISCUSSION

Based on the above-described kinetic models and numerical method, the numerical results of the near-cathode region and the cathode body for current-driven DC atmospheric-pressure argon arc discharge are reported and discussed in this section.

A. Distributions of plasma parameters in the near-cathode region

The distributions of typical kinetic parameters of arc plasma in the near-cathode region, including the electrostatic potential φ, number densities of ions and electrons (n_i and n_e), spatially averaged temperatures of ions and electrons (T_i and T_e), electron energy distribution function (EEDF), and ion energy distribution function (IEDF) are shown in Figs. 4–7 at different current densities of $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}.

Because of different physical mechanisms in different regions, the kinetic characteristics of arc plasma present obvious differences between the sheath and pre-sheath. It can be seen that there are

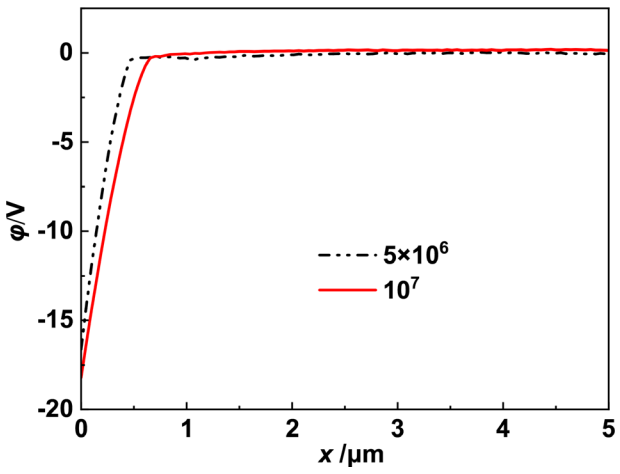


FIG. 4. Distribution of electric potential in the near-cathode region at $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}.

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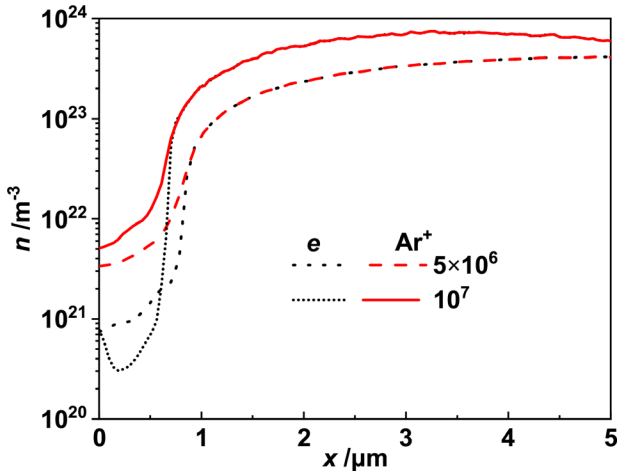


FIG. 5. Distribution of particles number density in the near-cathode region at $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}.

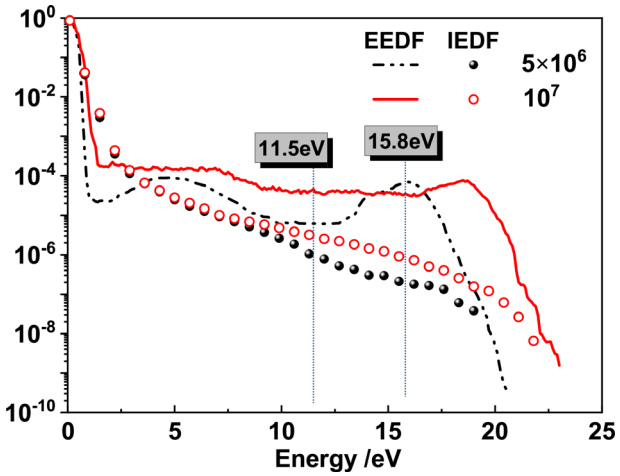


FIG. 7. EEDF and IEDF at $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}.

significant changes in plasma parameters in the space-charge sheath. Since both diffusion and drift of charged particles to the cathode begin to play more pronounced parts, the ionization and quasi-neutral equilibrium are disrupted by the bipolar diffusion and the convective transport of charged particles. Here, $n_e \neq n_i$ and $T_e \neq T_i$, which means the arc plasma greatly deviates from the chemical and thermal equilibrium in the sheath. In Fig. 4, the potential fall in the near-cathode region is about 16.62 and 18.21 V for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, which accords closely with typical characteristics of arc plasma with low voltage and high current.[7–15] The potential fall is largely concentrated in the sheath, while it is substantially lower outside the sheath (i.e., in the quasi-neutral plasma) and the change with x is not visible on the graph. This can be seen mathematically from Eq. (1) that the second derivative of the electric potential is directly related to the separation of charges.

The number density of ions (n_i) is much larger than that of electrons (n_e) due to the larger mass and lower mobility of ions compared with those of electrons when they are driven by the same electric field inside the cathode sheath. The sheath provides a thin positive space charge layer that limits the electron's losses to the boundary. The number density of charged particles in this region increases with a decrease in x for $j_{total} = 5 \times 10^6$ A m^{-2} in Fig. 5. A difference is that the number density of electrons under conditions of $j_{total} = 10^7$ A m^{-2} presents first decrease and then increase trend with a minimum value at $x = 0.2$ μm. A consequence of the above-described difference originates in the increase in the total current density to cause more emission electrons. Similar phenomena were also described in Ref. 25. To give an idea of an "edge" of the sheath, the thickness of the cathode sheath (δ) is evaluated as the region from the cathode to the position at which the separation of charges reaches 1%. $\delta = 1.1$ and 0.94 μm for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, can be determined, which is less than the results of Ref. 42, i.e., ~0.02 mm. That is because a different thoriated-tungsten cathode is applied and a collision-free sheath is assumed in the model of Ref. 42.

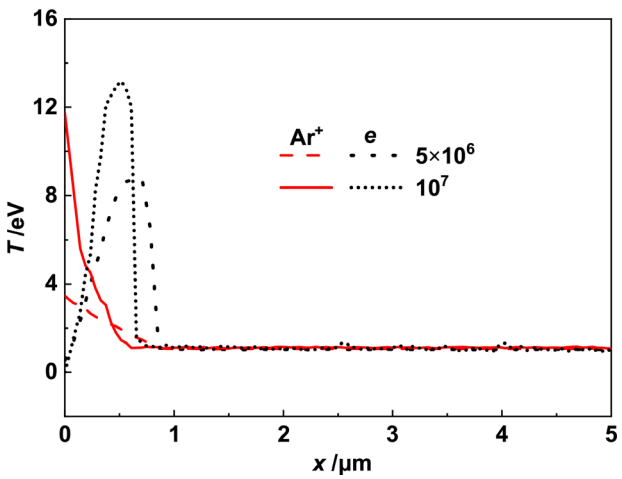


FIG. 6. Distribution of spatially averaged temperature of charged particles in the near-cathode region at $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}.

Figure 6 shows the variation of spatially averaged temperature (T_i and T_e) across the near-cathode region. The averaged temperature here is statistically obtained considering both the total energy and number density of charged particles. It is observed that T_e increases dramatically to reach a maximum and then decreases dramatically in the sheath. This can be understood as follows: (1) the electrons are continuously accelerated by the electric field to accumulate energy, explaining the rise in the temperature profiles at first; (2) the high-energy electrons can collide with gas atoms by excitation or ionization when their energy is above the excitation or ionization threshold, thereby losing their energy, leading to a sharp drop in the temperature profiles. These peaks are 8.84 eV and 13.19 eV for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, respectively. On the contrary, T_i varies monotonically and presents a sharp increase as x decreases. That is because ion energy mostly comes from the acceleration of the local electric field. Thus, the maximum values of T_i appear in the cathode of 3.46 and 11.75 eV for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, respectively, larger than the cathode temperature. At the right boundary of the pre-sheath, the arc plasma is nearly in local thermodynamic equilibrium (LTE) and local

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chemical equilibrium (LCE), i.e., $T_i \approx T_e$ and $n_i \approx n_e$. The spatially averaged temperature at the boundary of the near-cathode region is about ~1 eV, which should be high enough to maintain the unified ionization degree. The number densities of charged particles are about 10^{23}–10^{24} m^{-3} (seen in Fig. 5). Both of them are consistent with characteristic values of the charged particles inside the pre-sheath by numerical calculations in Refs. 43–45.

To analyze the particle kinetics in more detail, the spatially averaged EEDF and IEDF are shown in Fig. 7. The profiles of EEDF and IEDF deviate from Maxwell's distribution, which is consistent with the previous results.[46,47] EEDFs show multi-peak group distribution, and IEDFs show a two-energy group distribution, the same as Refs. 33 and 40 while this phenomenon becomes less obvious at a larger current density. In addition, with the increase in total current density, there are more charged particles with the same energy.

In the low energy region, i.e., <1 eV for electrons and <3 eV for ions, the population of particles almost linearly decreases with increasing energy, which is mainly attributed to the majority of particles in the pre-sheath with lower energy. This collisional heating enhances the low-energy ions heating, so ions efficiently escape from the low energy region of the IEDF to the high energy region of IEDF. The effects of acceleration by the electric field increase the population of high-energy electrons. It is easy to see that the lower boundary (i.e., 12.6 and 13 eV for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}) of the first high-energy group falls into the energy region between the excitation (11.5 eV) and ionization (15.8 eV) thresholds of argon because the electrons will lose their energy near these energy values. The same phenomenon is observed in Ref. 40. To sustain a stable arc discharge, the electrons heated within the cathode sheath must produce enough electron–ion pairs by ionization to compensate for the loss of the electrons and ions into the arc column. This means that some of the electrons must be heated in the cathode sheath to be above the ionization threshold. Similarly, the produced ions in the sheath are also accelerated by the electric field to cause the energy of part ions higher than 15.8 eV. The electrons and ions in the high-energy tails can significantly influence the discharge process. The maximum of ion energy is about 19 and 22 eV for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, respectively. In the high energy region (>16 and 20 eV for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}), the population of electrons decreases with the increase in energy and the maximum of electron energy is about 20.5 and 23 eV for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, respectively.

B. Energy transport of arc plasma–cathode and cathode–cold gas

To analyze the energy and current transport of arc plasma–cathode and cathode–cold gas, the temperature distributions of the cathode surface and current densities of the emitted electron are illustrated in Fig. 8. The distributions of heat flux are illustrated in Fig. 9, including the energy gain (i.e., P_{Joule} and $q_i + q_e$) and energy loss with cold gas (i.e., q_{rad} and q_{nc}), for the case with different total current densities, $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}.

As described above, there is almost no plasma generated when the cathode temperature is less than 2000 K. Therefore, the current density and heat flux of the charged particle are set to zero before the breakpoint for this case. The positions where there is no plasma current collection are at 8.7 and 10 mm for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, respectively. It is found in Fig. 8 that the cathode temperature increases

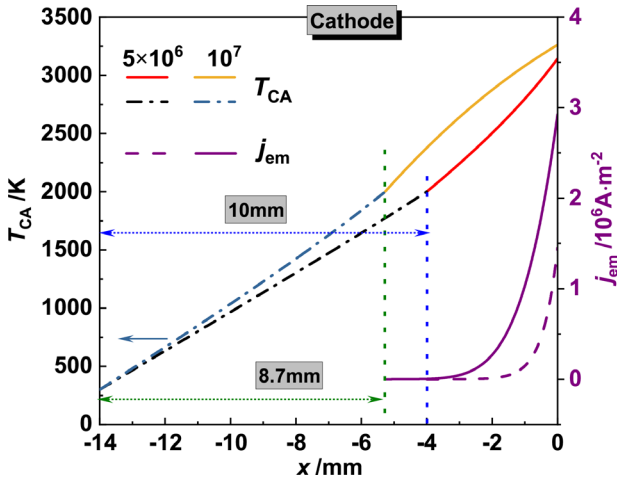


FIG. 8. Distributions of the cathode temperature and emission current density along the surface of a tungsten cathode.

almost linearly at $T_{CA} < 2000$ K and increases nonlinearly at $T_{CA} > 2000$ K. As could be expected, the hottest point of the cathode is the front face of the cathode, i.e., 3120 and 3260 K for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, respectively. Since the current density of emitted electrons j_{em} is proportional to the square of the cathode temperature [seen in Eq. (11)], j_{em} of $j_{total} = 10^7$ A m^{-2} is larger than $j_{total} = 5 \times 10^6$ A m^{-2}. Meanwhile, j_{em} increases exponentially with the increase in the cathode temperature and achieves maximum value on the right boundary of the cathode with $j_{em} = 1.45 \times 10^6$ A m^{-2} and $j_{em} = 2.94 \times 10^6$ A m^{-2} for $j_{total} = 5 \times 10^6$ and 10^7 A m^{-2}, respectively. As shown in Figs. 5 and 6, there is an intensive ionization process in the pre-sheath region and an intensive electric field near the cathode to form the ionic sheath, causing the emitted current density and the ionic current density are the main components of the imposed total current density.

Joule heating P_{Joule} by the external circuit and q_i+q_e from charged particles deposited into the cathode are the main components of energy gaining, while radiation is an important cause of energy loss compared with natural convection heat transfer, as can be seen in Fig. 9. Particle deposition plays a decisive role to supply sufficient energy to heat the surface to temperatures necessary for strong thermionic emission while Joule heating and radiation are almost balanced. The densities of energy fluxes of Joule heat, natural convection, and radiation show nonlinear distribution monotonically increasing because of the temperature distribution of the cathode surface.

C. Effects of different total current densities on plasma characteristics

Electric field and charge density at $x = 0$, i.e., E_0 and σ_0, and the maximum particle number density n_{max} for different total current densities (10^6–10^7 A m^{-2}) are shown in Fig. 10. The maximum value of spatially averaged particle temperature T_{max} and the thickness of the sheath for different total current densities are presented in Fig. 11. Obviously, E_0, σ_0, n_{max}, and T_{max} increase monotonically with the increase in j_{total}. However, the thickness of the sheath presents the opposite case. As the total current density tends to higher values, the thickness of the sheath slightly changes with a smaller value below 1 μm. These results are

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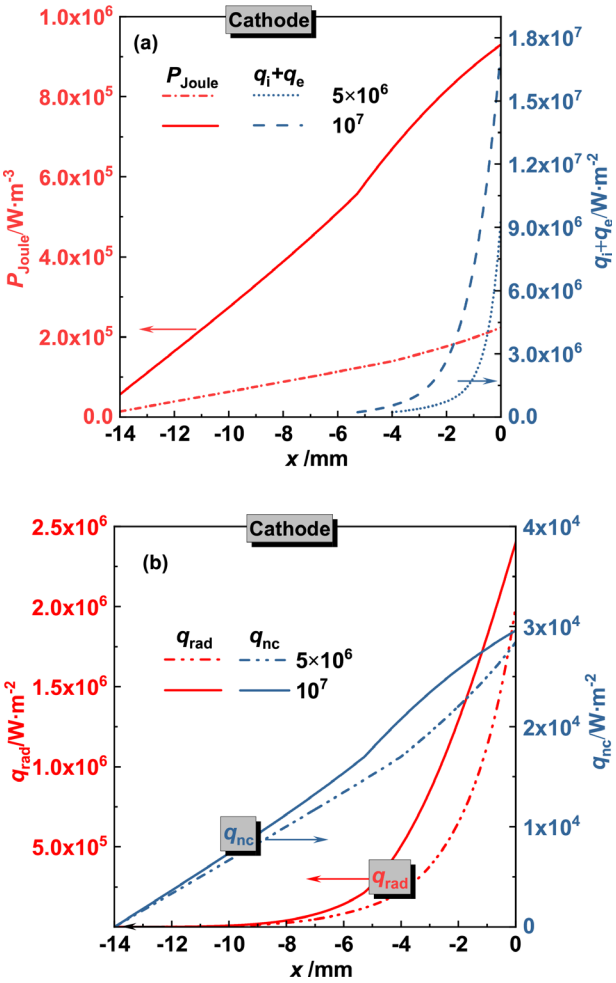


FIG. 9. Distributions of the flux densities along the cathode surface. (a) Flux densities of the energy gain (i.e., P_{Joule} and $q_i + q_e$). (b) Flux densities of the energy loss with cold gas (i.e., q_{rad} and q_{nc}).

understandable and can be explained as follows. The electric power deposited in the near-cathode region increases with the increase in the total current. More electric power carried by charged particles is transported to the cathode to enhance the electron emission. This greater emission intensity leads to more ions through collisions. This growth of positive space charge subsequently increases the electric field at the cathode, which also increases emission. In essence, a feedback-like mechanism occurs where increased emission leads to more ionization, which increases the emission further. Thus, the variation in n_{max} is in an exponential form. The increase in the number density of particles allows more ions to move to the sheath region to enhance the electric field. In addition, a stronger electric field also further increases the kinetic energy of the ions and electrons, which increases their penetration depth, thus leading to a decrease in the thickness of the sheath. Thus, the numbers of ions deposited into the cathode surface also increase, leading to an increase in the cathode temperature. Since thermionic emission is dependent on not only the electric field but also the cathode temperature [seen in Eq. (11)], a similar feedback effect occurs as the cathode

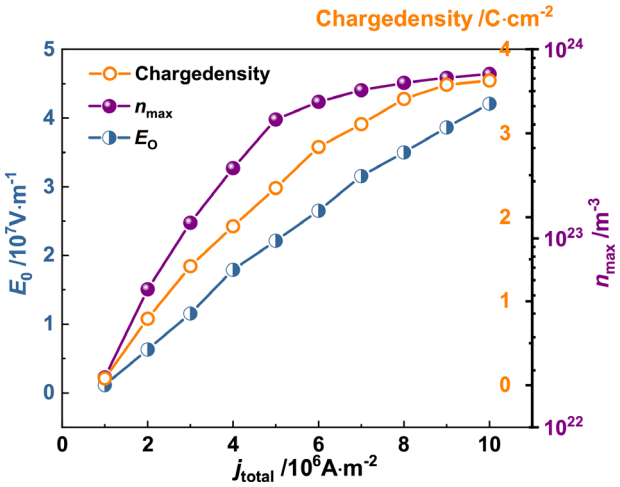


FIG. 10. Electric field and charge density at $x = 0$, and maximum particle number density for different total current densities.

temperature increases. That is, the electrons inside the cathode can receive more energy to be emitted, and the ionization probability increases due to the elevated electrons density and energy at higher cathode temperature.

The temperature of the cathode at $x = 0$, T_{CA0}, as a function of the total current density is represented in Fig. 12. In this case, the length of the tungsten cathode is 10 mm, and the radius is 2 mm according to the physical model by Benilov[48] In their work, a 1D planar model is developed for describing the process that current transfers to a planar cathode through a planar near-cathode region in DC argon arc discharge at $p = 1$ atm. Since the effects of Joule heat generation and energy loss inside the cathode body are ignored in this model, the temperature of the cathode T_{CA0} given by Ref. 48 is a bit higher than that of the present work under the same j_{total}. It can be seen that the cathode temperatures present a nonlinear increase with the increase in the current density in both the present work and Benilov's

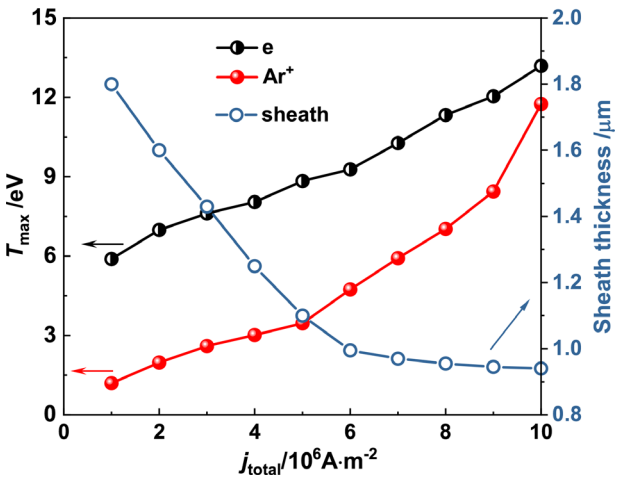


FIG. 11. Maximum of spatially averaged particle temperature and thickness of sheath for different total current densities.

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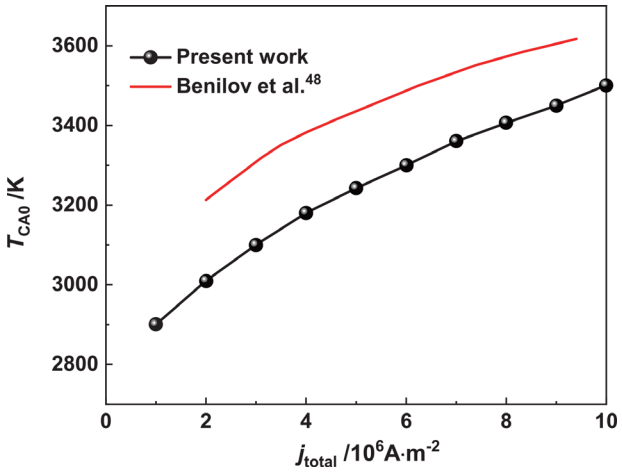


FIG. 12. Cathoder temperature at $x = 0$, T_{CA0}, for different current densities of a tungsten cathode with a height of 10 mm and radius of 2 mm at $p = 1$ atm. Line and symbol: present work. Line: the model of Benilov in Ref. 48.

work.[48] The characteristic curves of electric potential at $x = 0$, φ_0, and total current densities for a fixed cathode temperature $T_{CA0} = 3500$ K at $p = 1$ bar are illustrated in Fig. 13. Here, the longitudinal coordinate axis is the absolute value of φ_0. In the framework of Refs. 25 and 35, hydrodynamic (diffusion) descriptions of the motion of the ions and the electrons are employed to describe the particle transport in the near-cathode region based on the assumption of a collision-free sheath. In the present work, the collision between charged particles is determined self-consistently based on kinetic energy and distances. The electrons emitted by the cathode surface can receive a substantial energy supply through acceleration by the electric field to make the ionization possible in the space-charge sheath. Consequently, there is a certain quantitative difference between the characteristic curve of this

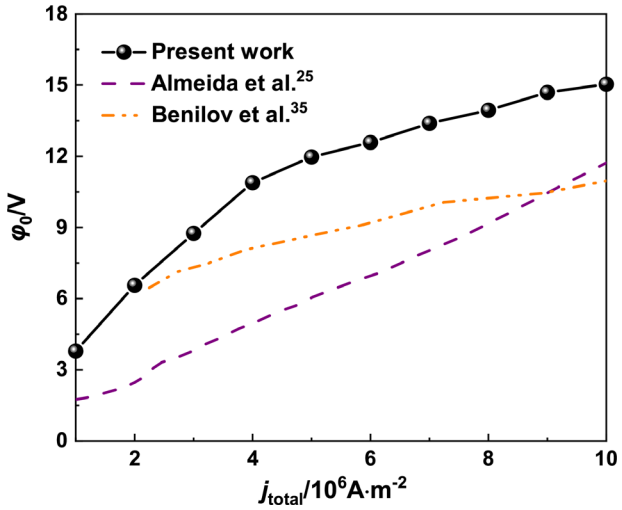


FIG. 13. Characteristic curves of electric potential at $x = 0$ φ_0 and j_{total} at a fixed cathode temperature $T_{CA0} = 3500$ K at $p = 1$ bar. Solid: present work. Dash: the model of Almeida *et al.* in Ref. 25. Dot-dot-dash: the model of Benilov *et al.* in Ref. 35.

work and that of Ref. 35. In addition, all the results indicate that the electric potential increases with the increase in the total current density at a constant cathode temperature. Since the cathode emission mechanisms are strong enough to sustain high currents, φ_0 shows little values of 3–18 V as shown in Fig. 13.

IV. CONCLUSIONS

In this paper, 1D unified model combining the near-cathode region and the cathode is developed to study the plasma transport characteristics and the arc plasma–cathode and cathode–cold gas interactions for the atmospheric pressure arc discharge driven by a current source. The main conclusions are as follows.

(1) This point of view of the collision-dominated near-cathode region is supported, and there is an intensive ionization process in this region. Meanwhile, EEDFs show a multi-peak group distribution and IEDFs show a two-energy group distribution while this phenomenon becomes less obvious with a larger current density.
(2) For j_{total} of 10^6–10^7 A m^{-2}, the calculated number densities are in the order of 10^{22}–10^{24} m^{-3} and the spatially averaged temperatures are several eV. The potential fall in the near-cathode region is lower than 20 V and the electric field is $\sim 10^7$ V m^{-1}. The range of typical parameters is consistent with the arc discharge characteristics in qualitative and quantitative terms.
(3) Joule heating by the external circuit and charged particles deposited into the cathode are the main components of energy gaining. Particle deposition plays a decisive role in supplying sufficient energy to heat the surface to temperatures necessary for thermionic emission. Energy loss by radiation is more prominent compared with natural convection.
(4) The total current density has more significant effects on the kinetic characteristics of arc plasma. The thickness of the space-charge sheath decreases nonlinearly with the increase in the total current densities. A feedback-like mechanism occurs in which the enhanced emission leads to more ionization and then increases the emission further.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Li Sun: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review and editing (equal). **Xian-Pin Sun:** Writing – review and editing (equal). **Bi-Ying Guo:** Writing – review and editing (equal). **Wen Zhou:** Writing – review and editing (equal). **Zeng-Yao Li:** Conceptualization (equal); Writing – original draft (equal); Writing – review and editing (equal).

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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